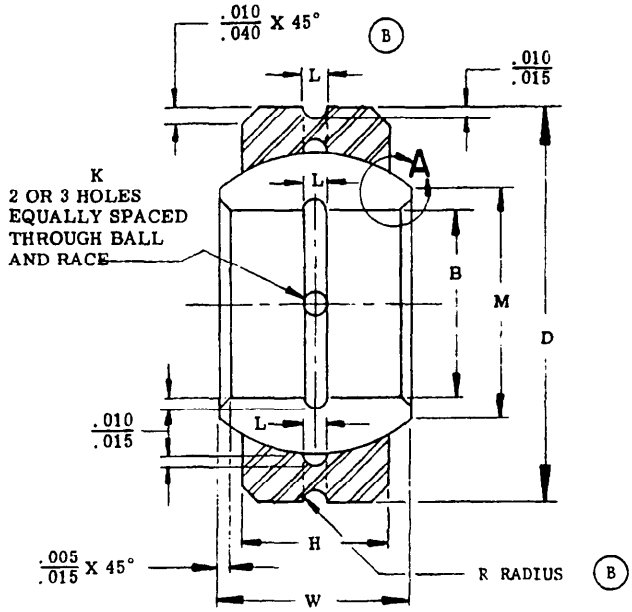
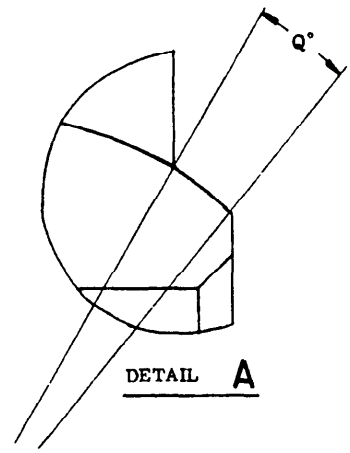
FED. SUP CLASS
3120





NOTES:

(B) 1. MATERIAL: BALL - 52100, 440C AMS 5630.
RACE - CODES: "S" ALLOY STEEL, MIL-S-5000, MIL-S-6758 OR MIL-S-6050
"B" ALUMINUM BRONZE (AMPCO 15) OR ALUMINUM BRONZE PER QQ-C-465 - EXCEPT THAT CHEMICAL COMPOSITION SHALL BE:

ELEMENT	Cu	Al	Fe	Ni	Si	Mn	Sn	Zn	SUM OF NAMED ELEMENTS MIN
PERCENT	80.0 to 93.0	6.5 to 11.0	4.0 1/	1.0 1,2/	2.2 1,2/	1.5 1/	.60 1/	1.0 1/	99.5

1/ MAXIMUM

2/ WHEN BOTH SILICON AND NICKEL ARE PRESENT ONLY ONE SHALL BE IN EXCESS OF 0.25 PERCENT.

2. FINISH: BALL - CHROME PLATE CLASS 2, .0002 MINIMUM ON SPHERICAL SURFACES, .0005 ON FACES.
RACE - CADMIUM PLATE PER QQ-P-416, TYPE 1, CLASS 2.

3. HARDNESS: BALL - Rc 56 MIN.
RACE - Rc 27-36 ALLOY STEEL.

4. LUBRICATION: PREPACKED WITH MIL-G-21164.

5. SURFACE TEXTURE: SPHERICAL SURFACE OF BALL 8 RHR; BORE, BALL FACE AND OUTER FACE O.D. 32 RHR. ALL OTHER SURFACES 125 RHR. IN ACCORDANCE WITH ANSI B46.1.

6. TEMPERATURE RANGE: -65°F TO +250°F.

7. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .010, ANGLES ± 1/2°.

8. BREAK ALL SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

EXAMPLE OF PART NUMBER:

MS21155 X XX
- XX: BORE CODE IN MULTIPLES OF 1/16 INCHES.
- X: RACE MATERIAL CODE
- MS21155: BASIC PART NO.

(B) 9. INTERNAL CLEARANCE: RADIAL - 0.0005 TO 0.0020 INCH.
AXIAL - 0.010 MAXIMUM.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(B) DENOTES CHANGES

APPROVED 17 Mar 1969 REVISED (A) 9 JUN 72 (B) 15 SEPT 80

P.A. NAVY - AS Other Cust USAF - 11 ARMY - AV	TITLE: BEARING, PLAIN, SELF ALIGNING	MILITARY STANDARD MS21155
PROCUREMENT SPECIFICATION: MIL-B-8976	SUPERSEDES: MS21155(ASG)	SHEET 1 OF 3

DD FORM 1 MAR 72 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 17709

USER SYMBOLS:
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REVIEWER SYMBOLS:

USER SYMBOLS:

DD FORM 1000, MAR 72 **672-1** (Coordinated)

P.A. NAVY - AS

Other Cust

USAF - 11

ARMY - AV

PROCUREMENT SPECIFICATION MIL-B-8976

SUPERSEDES: MS21155(ASG)

TITLE

BEARING, PLAIN, SELF ALIGNING

MILITARY STANDARD

MS21155

SHEET 2 OF 2

DASH NO.	B +.0000 -.0005	D +.0000 -.0005	H ±.005	K	L ±.005 (a)	M	Q° MIN	W +.000 -.002	ALLOY STEEL RADIAL STATIC LIMIT LOAD(b)	ALLOY STEEL AXIAL STATIC LIMIT LOAD	ALUMINUM BRONZE RADIAL STATIC LIMIT LOAD(b)	ALUMINUM BRONZE AXIAL STATIC LIMIT LOAD
3	.1900	.5625	.218	.047	.062	.293	10	.281	4600	2100	2800	850
4	.2500	.6562	.250			.364		.343	7080	2800	4300	1100
5	.3125	.7500	.281	.062	.078	.419		.375	8500	3550	5200	1400
6	.3750	.8125	.312			.475	9	.406	11050	4400	6750	1760
7	.4375	.9062	.343			.530	8	.437	13900	5400	8500	2150
8	.5000	1.0000	.390			.600		.500	18850	7050	11500	2800
9	.5625	1.0937	.437			.670		.562	25500	8900	15600	3550
10	.6250	1.1875	.500	.078	.093	.739		.625	31950	11700	19500	4650
12	.7500	1.4375	.593			.920		.750	46750	16500	28500	6575
14	.8750	1.5625	.703			.980		.875	62750	23300	38300	9300
16	1.0000	1.7500	.797			1.118	9	1.000	83350	30000	51000	12000

(a) L DIMENSIONS AND HOLE DIAMETER TO BE MEASURED BEFORE BEARING ASSEMBLY BUT SWAGING SHALL NOT RESTRICT GREASE FLOW.

(B) (b) NOTE: THESE BEARINGS MAY EXPERIENCE SOME METAL YIELDING AT THE RADIAL STATIC LIMIT LOADS LISTED.

(B)

DASH NO.	R RADIUS (REF)	BALL DIA. MAX. ALLOY STEEL	BALL DIA. MAX. ALUM. BRONZE
-3	.045	.407	.438
-4		.501	.501
-5	.065	.563	.594
-6		.657	.657
-7		.719	.719
-8		.814	.814
-9		.907	.907
-10	.088	.970	1.001
-12		1.188	1.251
-14		1.313	1.376
-16		1.501	1.563

FED. SUP CLASS 3120

APPROVED 17 Mar 1969 REVISED (B) FOR CHANGES SEE SHEETS 1 AND 2.

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